E.G.S.PILLAY ARTS &SCIENCE COLLEGE(AUTONOMOUS)
NAGAPATTINAM–611002.TAMILNADU
(Accredited by NAAC With"A" Grade)
(MBA – Approved by AICTE, New Delhi)
(Affiliated to Bharathidasan University, Tiruchirappalli–24)

# M.Sc- COMPUTER SCIENCE

## SYLLABUS

**Choice Based Credit System- Learning Outcomes Based Curriculum Framework (CBCS-LOCF)**

**(Applicable to the candidate admitted from the academic year 2025–2026 onwards)**

# M.Sc-MasterofComputerScience

**Choice Based Credit System –Learning Outcomes Based Curriculum Framework (CBCS- LOCF)**
**(Applicable to the candidate admitted from the academic year 2025–2026 onwards)**

| Sem. | Courses | Title | Ins. Hrs | Credits | Exam Hours | MaximumMarks | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Int. | Ext. | Total |
| I | CoreCourse–I(CC) | AdvancedJavaProgramming | 6 | 4 | 3 | 25 | 75 | 100 |
| | CoreCourse–II(CC) | DesignandAnalysisofAlgorithms | 6 | 4 | 3 | 25 | 75 | 100 |
| | CoreCourse–III(CC) | Advanced WebTechnologies | 6 | 4 | 3 | 25 | 75 | 100 |
| | ElectiveCourse–I(EC) | **A.** Mathematical Foundation for Computer Science<br>**B.** Statistical Computing | 6 | 4 | 3 | 25 | 75 | 100 |
| | Core Practical–I(CP) | AdvancedJavaProgrammingLab | 3 | 3 | 3 | 50 | 50 | 100 |
| | CorePractical–II(CP) | Advanced WebTechnologiesLab | 3 | 3 | 3 | 50 | 50 | 100 |
| | **Total** | | **30** | **22** | | | | **600** |

| Sem. | Courses | Title | Ins. Hrs | Credits | Exam Hours | MaximumMarks | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Int. | Ext. | Total |
| II | CoreCourse–IV(CC) | Advanced Database systems | 6 | 4 | 3 | 25 | 75 | 100 |
| | Core Course–V(CC) | Problem solving using Python and R | 5 | 4 | 3 | 25 | 75 | 100 |
| | Core Course–VI(CCC) | Mobile Application Development | 5 | 4 | 3 | 25 | 75 | 100 |
| | Core Practical–III(CP) | Advanced Database systems lab | 3 | 3 | 3 | 50 | 50 | 100 |
| | Core Practical–IV(CP) | Problem solving using Python and RLab | 3 | 3 | 3 | 50 | 50 | 100 |
| | Elective Course–II(EC) | **A.** Internet of Things<br>**B.** Artificial Intelligence | 5 | 4 | 3 | 25 | 75 | 100 |
| | Non-major Elective Course I | **A.** Cryptography<br>**B.** Fundamentals of BIGDATA | 3 | 2 | 3 | 25 | 75 | 100 |
| | | | **30** | **24** | | | | **700** |

## M.Sc-Master of Computer Science

**Choice Based Credit System –Learning Outcomes Based Curriculum Framework (CBCS- LOCF)**
**(Applicable to the candidate admitted from the academic year 2025–2026 onwards)**

| Sem. | Courses | Title | Ins. Hrs | Credits | Exam Hours | Maximum Marks | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Int. | Ext. | Total |
| III | Core Course–VII(CC) | Big Data Analytics | 6 | 4 | 3 | 25 | 75 | 100 |
| | Core Course–VIII(CC) | Compiler Design | 5 | 4 | 3 | 25 | 75 | 100 |
| | Core Practical–V(CP) | Big Data Analytics lab | 3 | 3 | 3 | 50 | 50 | 100 |
| | Elective Course–III(EC) | **A.** Multicore Computer Architecture.<br>**B.** Agile Technologies | 5 | 4 | 3 | 25 | 75 | 100 |
| | Core Course- IX(CCC) | Cryptography and Network Security. | 5 | 4 | 3 | 25 | 75 | 100 |
| | Core Practical-VI(CP) | Cryptography and Network Security Lab | 3 | 3 | 3 | 50 | 50 | 100 |
| | Non-Major Elective Course II | **A.** DHTML and XML<br>**B.** Cyberlaw and Crimes | 3 | 2 | 3 | 25 | 75 | 100 |
| | INTERNSHIP | Internship<br>i. Report of the study (80)<br>ii. Viva voce(20) | - | 3 | - | 20 | 80 | 100 |
| | **Total** | | **30** | **27** | - | - | - | **800** |

| Sem. | Courses | Title | Ins. Hrs | Credits | Exam Hours | Maximum Marks | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Int. | Ext. | Total |
| IV | Core Course–X(CC) | Block Chain Technology | 6 | 4 | 3 | 25 | 75 | 100 |
| | Core Course–XI(CC) | Machine Learning Technologies | 6 | 4 | 3 | 25 | 75 | 100 |
| | SOC-Skill Oriented Course | Technology Innovation and Sustainable Enterprise | 4 | 4 | 3 | 25 | 75 | 100 |
| | PROJECT | Project<br>i. Dissertation(80)<br>ii.Viva voce (20) | 14 | 5 | - | 20 | 80 | 100 |
| | | **TOTAL** | **30** | **17** | - | - | - | **400** |
| | **GRAND TOTAL** | | **120** | **90** | | | | **2500** |



20/05/2026
DR.M.LALLI, MCA.,M.Phil.,Ph.D.,
Associate Professor
Department of Computer Science,
Bharathidasan University,
Trichy - 620 023.

20/5/26
Dr. A. ADHISELVAM, Ph.D.
PROFESSOR AND HEAD
Department of Information Technology
Dr. N.G.P. Arts and Science College
Coimbatore - 641 048

20/5/26
Dr.P.PANIMALAR, M.C.A.,M.Phil.,S.E.T.,Ph.D.,
ASSOCIATE PROFESSOR OF COMPUTER SCIENCE
A.V.C. COLLEGE (AUTONOMOUS)
MANNAMPANDAL MAYILADUTHURAI-609 305.
Mobile No: 94437 43090

## Summary of Curriculum Structure of M.SC- CS Programme

| Sl. No. | Types of Courses | No. of Courses | No. of Credits | Marks |
|---|---|---|---|---|
| 1. | Core Course | 11 | 44 | 1100 |
| 2. | Core Practical | 6 | 18 | 600 |
| 3. | Elective Courses | 3 | 12 | 300 |
| 4. | Non- Major Elective Courses | 2 | 4 | 200 |
| 5. | Project | 1 | 5 | 100 |
| 6. | Skill Oriented Course | 1 | 4 | 100 |
| 7. | Internship(15 to 30 days) | 1 | 3 | 100 |
| **Total** | | **25** | **90** | **2500** |

20/05/2026

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Mobile No: 94437 43090

**Programme Outcomes:**

- Torecognizethevalueoflifelonglearningandtheroleofscienceinsociety.
- To identify the contemporary issues and demonstrate the scientific knowledge through various processes to arrive at solutions
- To collect, assess and make use of the information in both theoretical and experimental science
- To apply advanced theoretical and/or experimental methods, including the use of laboratory techniques, numerical methods and computer simulations in various fields and practical problems
- To disseminate the science information effectively both in written or by orally before any audience.

**Programme Specific Outcomes:**

- To apply the knowledge of Computer Science to identify, analyze problems and to provide effective solution in the area of Computing
- To develop skills to excel in the fields of Computer Science and its enabled services in various fields of Government and Private sectors
- To employ modern computer languages and applications for their successful career, and to create platforms to become an entrepreneur
- To instill ethical responsibilities, human and professional values and make their contribution to the society.

| FIRST YEAR | CORECOURSE–I | COURSE CODE:P25CSCC11 |
|---|---|---|
| SEMESTER-1 | ADVANCEDJAVAPROGRAMMING | CREDITS: 4 |

**COURSE OBJECTIVES:**

- TogainknowledgeofObjectOrientedProgrammingConceptinJava.
- TounderstandusagesofStringfunctionsinJava.
- Tofamiliarizewiththeappletandswing.
- Tograsptheconceptson JavaBeans.
- TocomprehendtheconnectionbetweenRelationalDatabaseandJava.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | An Overview of Java: Object Oriented Programming- Data Types, Variables, and Arrays: Primitive Types- Literals Variables - Type Conversion and Casting- Arrays-Operators: Control Statements-Classes and Methods – Inheritance- Exception Handling. | 20 |
| II | String Handling: The String Constructors -String Length- SpecialString Operations –CharacterExtraction- String Comparison - Searching Strings - Modifying a String – Input/Output: The I/O Classes and Interfaces – File - Byte Streams - Character Streams. | 20 |
| III | The Applet Class: Basic Architecture - Applet Skeleton - Display methods - Status Window – Passing Parameters. Introducing GUI Programming with Swing– Introducing Swing - Swing Is Built on the AWT- Two Key Swing Features - The MVC Connection - Components and Containers - The Swing Packages - A Simple Swing Application - Exploring Swing. | 20 |
| IV | JavaBeans:Introduction-AdvantagesofBeans –Introspection-TheJavaBeansAPI–ABeanExample.Servlets:LifeCycleSimpleServlet-ServletAPI-Packages-Cookiessessiontracking. | 15 |
| V | NetworkProgramming:WorkingwithURLs-WorkingwithSockets-RemoteMethodInvocation.IntroductiontoDatabaseManagementSystems-Tables ,Rows andColumns-IntroductiontotheSQL SELECT Statement - Inserting Rows - Updating and Deleting Existing Rows - Creating and Deleting Tables Creatinga New Database with JDBC-Scrollable Result Sets. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understand the Object Oriented Program including classes and methods; inheritance and exception handling. | K2 |
| CO2 | Complete comprehension of String functions and I/O Streams. | K3 |
| CO3 | Creation of graphical representation using Applet. | K6 |
| CO4 | Application of Servlet for designing Web based applications. | K6 |
| CO5 | Usage of JDBC connectivity and implementation of the concept to get desired results from database. | K3 |

**Text Books:**

1. HerbertSchildt,"JavatheCompleteReference",10th Edition,McGrawHillPublishingCompanyLtd,New Delhi, 2023.
2. Tony Goddis,"StartingoutwithJavafrom Control StructuresThroughObjects"6th Edition, Pearson Education Limited, 2022.

**Referencebooks:**

1. HerbertSchildt,DaleSkrien,"JavaFundamentals –AComprehensive introduction",TMGH Publishing Company Ltd, New Delhi, 2024.
2. JohnDean,RaymondDean,"IntroductiontoProgrammingwithJAVA–ProblemSolving Approach", TMGH Publishing Company Ltd, New Delhi,2021.

**WebResource link:**

https://github.com/JJBRT/advanced-java-tutorials
https://www.reddit.com/r/learnjava
https://www.geeksforgeeks.org/advance-java/advanced-java/

| FIRST YEAR | CORECOURSE–II | COURSE CODE:P25CSCC12 |
|---|---|---|
| SEMESTER-1 | DESIGN AND ANALYSISOFALGORITHMS | CREDITS:4 |

**COURSE OBJECTIVES:**

- To study the concepts of algorithms
- To analysis of algorithms using divide and conquer,
- To analysis of greedy method and dynamic Programming
- To analysis also backtracking, and branch and bound techniques.
- To study about Elementary Data Structures

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | Introduction: Algorithm Definition–Algorithm Specification–Performance Analysis. Elementary Data Structures: Stacks and Queues – Trees – Dictionaries – Priority Queues–Graphs. | 20 |
| II | Divide and Conquer: The General Method–Defective Chessboard–Binary Search–Finding the Maximum and Minimum – Merge Sort – Quick Sort – Selection - Stassen's Matrix Multiplication. | 20 |
| III | The Greedy Method: General Method - Knapsack Problem - Tree Vertex Splitting – Job Sequencing With Deadlines - Minimum Cost Spanning Trees - Optimal Storage On Tapes – Optimal Merge Patterns- Single Source Shortest Paths. | 20 |
| IV | Dynamic Programming: The General Method–Multistage Graphs–All-Pairs Shortest Paths–Single-Source Shortest Paths-Optimal Binary Search Trees-The Traveling Salesperson Problem. Basic Traversal and Search Techniques: Techniques for Binary Trees – Techniques for Graphs – Bi connected Components and DFS. | 15 |
| V | Back tracking: The General Method –The 8-Queens Problem – Sum of Subsets – Graph Coloring –Hamiltonian Cycles – Knapsack Problem Branch and Bound: The General Method - 0/1 Knapsack Problem. | 15 |

**COURSE OUTCOMES:**
Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| C01 | Able to analyze the different algorithm design techniques for a given problem. | K4 |
| C02 | Able to design algorithms for various computing problems. | K6 |
| C03 | Able to argue the correctness of algorithms using inductive proofs and invariants. | K5 |
| C04 | Argue the correctness of algorithms using inductive proofs and invariants. | K5 |
| C05 | Analyze worst-case running times of algorithms using asymptotic analysis. | K4 |

**TEXTBOOKS:**

1. AnanyLevitin"IntroductiontotheDesignandAnalysisofAlgorithms"PearsonEducation2024.
2. Introduction to Algorithms, byThomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, 4th edition 2024.

**REFERENCES BOOKS:**

1. ThomasH.Cormen,CharlesE.Leiserson,RonaldL.Rivest,"Introductionto algorithms",PrenticeHall2020.
2. S.K.Basu,"DesignmethodsandAnalysisofAlgorithms",PrenticeHall,2022.

**WEBRESOURCE LINK:**

1.https://xlinux.nist.gov/dads/HTML/?utm_source
2.https://www.reddit.com/r/webdev/comments/iito08/?utm_source=chatgpt.com
3.https://www.geeksforgeeks.org/dsa/dsa/

| FIRST YEAR | CORECOURSE–III | COURSECODE: P25CSCC13 |
|---|---|---|
| SEMESTER-1 | ADVANCED WEB TECHNOLOGIES | CREDITS:4 |

**COURSEOBJECTIVES:**

- To provide fundamental concept of Internet, JavaScript, XML, JSP,
- ASP with a viewtodevelopingprofessionalsoftwaredevelopmentskills.
- Tomakefamiliarwithclientserverarchitecture.
- Togetproject-basedexperienceneededforentryintowebapplication.
- To study about development careers.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **INTERNET BASICS:** Basic Concepts – Internet Domains – IP Address – TCP/IP Protocol – The WWW – The Telnet–Introduction to HTML: Web server- Web client /browser-Tags– Text Formatting–Lists–Tables–Linking Documents-Frames. | 20 |
| II | **JAVASCRIPT:** JavaScript in Web Pages – The Advantages of JavaScript –Writing JavaScript into HTML–Syntax–OperatorsandExpressions–Constructsandconditionalchecking – Functions – Placing text in a browser – Dialog Boxes – Form object's methods–Built in objects–user defined objects. | 20 |
| III | **XML:** Comparison with HTML – DTD – XML elements – Content creation –Attributes– Entities – XSL – XLINK – XPATH – XPOINTER – Namespaces –Applications – integrating XML with other applications. | 20 |
| IV | **JSP FUNDAMENTALS:** Basics – Directive basics – Page directive – The tag lib directive – The include directive–JSP Standard Actions–JavaBeans–Error Handling. | 15 |
| V | **ASP:** Introduction to ASP – Objects – Components – Working with HTMLforms– ConnectingtoMicrosoftSQLServer&MS–AccessDatabase–SQLstatementswith connection object – Working with record sets. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Design a webpage with Web form fundamental sand web control classes. | K6 |
| CO2 | Recognize the importance of validation control, cookies and session | K1 |
| CO3 | Apply the knowledge of ASP.NET object, ADO.NET data access and SQL to develop a client server model. | K3 |
| CO4 | Recognize the difference between Data list and Data grid controls in accessing data. | K1 |
| CO5 | Know the code optimization techniques; Understand the techniques of Internet, JavaScript, XML, JSP, ASP with a view to developing professional software development skills. | K1 |

**TEXTBOOKS:**

1. RobertW.Sebesta:ProgrammingtheWorldWideWeb,EighthEdition,Pearsoneducation,2024. **UNITS:**1,2,3,4
2. Dayley Brad, Dayley Brendan,"Angular S,JavaScript, and jQuery All in One",Sams Teach Yourself 1st Edition, Kindle Edition, 2024.

**REFERENCEBOOKS:**

1. M.Srinivasan:WebProgrammingBuildingInternetApplications,3rdEdition,WileyIndia,2020.
2. Jeffrey C. Jackson: Web Technologies-A Computer Science Perspective,PearsonEducation,7thImpression,2021.
3. Chris Bates: Web Technology Theory and Practice, PearsonEducation,2019. Raj Kamal: Internet and WebTechnologies, McGraw Hill Education.

**WEBRESOURCE LINK:**

1. https://jogindrakumar.com/top-free-websites-for-learning-web-development
2. https://www.geeksforgeeks.org/top-websites-to-learn-web-technologies
3. https://en.wikipedia.org/wiki/MDN_Web_Docs

| FIRST YEAR | ELECTIVE COURSE–IA | COURSE CODE: P25CSE1A |
|---|---|---|
| SEMESTER I | MATHEMATICAL FOUNDATION FOR COMPUTER SCIENCE | CREDITS:4 |

**COURSEOBJECTIVES**:

- Ability to apply mathematical logic to solve problems.
- Understand Propositions, tautologies, and inference rules.
- Understand gauss methods.
- Understand operations on sets.
- Able to formulate problems and apply the test of hypothesis.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | Determinants, inverse of matrix. System of equations, Linear transformation -rank and nullity, Consistency and inconsistency of linear system of equations, rank nullity theorem Echelon form of a matrix, and Row reduced echelon formof matrix. Eigen values and Eigen vectors. | 20 |
| II | Gauss elimination method, Gauss Jordan Method, JacobiMethod for solvinglinear systems. | 20 |
| III | Operations on sets, Venn Diagrams, Multi Sets, Binary Relations, EquivalenceRelations, Ordering Relations, Operations on Relations, Partial Orders. Statements and Notation, Connectives, Quantified Propositions, LogicalInferences, Methods of Proof of an Implication, First Order Logic and otherMethods of Proof, Rules of Inference for Quantified Propositions, Proof byMathematical Induction. | 20 |
| IV | Calculating Coefficients of Generating Functions, Recurrence Relations, SolvingRecurrence Relations by Substitution and Generating Functions, the Method ofCharacteristic Roots, Solutions of Inhomogeneous Recurrence Relations. | 15 |
| V | Random variables - discrete and continuous, probability functions, density anddistribution functions, mean and variance, special distributions (Binomial, Hypergeometric, Poisson, Uniform, exponential and normal). Testing of Hypothesis,Null and alternative hypothesis, level of significance, one-tailed and two tailedtests, tests for small samples- T-test, Chi-square test. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Compute **determinants** and **matrix inverses**, and use them to determine matrix invariability. | K3 |
| CO2 | Analyze and solve **linear systems** using Gauss elimination, Gauss–Jordan, and **Jacobi methods**, identifying consistency/inconsistency and evaluating convergence ties in numerical approaches. | K4 |
| CO3 | Apply set operations, interpret **Venn diagrams**, and handle **multisets** to model and solve counting and relational problems. | K3 |
| CO4 | List **recurrence relations**, derive and apply **generating functions**, and extract coefficients to characterize sequence behavior. | K1 |
| CO5 | Execute **hypothesis testing** including null/alternative formulation, significance levels, one-/two-tailed assessments, and proper test selection. | K2 |

**TEXTBOOKS:**

1.Kenneth H. Rosen, "Discrete Mathematics and Its Applications", 7th Ed, McGraw Hill, 2022.

2. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley India, 9th edition ,2021.

3. Bernard Kolman, Robert Busby and Sharon Cutler Ross, "DiscreteMathematical Structures for Computer Science", 6 th Ed, PHI, 2022.

**REFERENCES BOOKS:**

1. William Stein, "Elementary Number Theory: Primes, Congruences, andSecrets": A Computational Approach Springer, 2020.

2. Sipser, "Introduction to the Theory of Computation, CENGAGE Learning,2020.

3. Ernest Davis, "Linear Algebra and Probability for Computer ScienceApplications ", 1st Edition, CRC Press 2022.

**WEBRESOURCE LINK:**

1. https://www.tutorialspoint.com/mathematical foundation for computer science.
2. https://9meters.com/technology/phones/understandingmathematicalfoundation
3. https://en.wikipedia.org/wiki/mathematicalfundation

| FIRST YEAR | ELECTIVE COURSE–IB | COURSE CODE: P25CSE1B |
|---|---|---|
| SEMESTER-I | STATISTICAL COMPUTING | CREDITS:4 |

**COURSE OBJECTIVES**:

- **To introduce** the fundamentals of statistical computing and programming using statistical software such as R and Python.
- **To develop** a strong foundation in numerical methods and algorithms essential for statistical data analysis.
- **To enable** students to model, simulate, and interpret real-world data using statistical methods and machine learning techniques.
- **To impart** skills in effective data preprocessing, management, and visualization for insightful decision-making.
- **To provide** exposure to advanced tools, applications, and research trends in computational statistics and big data analytics.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **Foundations of Statistical Computing:**<br><br>Introduction-Statistical Software-Basics of Programming-Data Structures-Numerical Computing | 20 |
| II | **Numerical Methods in Statistics:**<br><br>Root-Finding Algorithms-Numerical Differentiation and Integration-Optimization Techniques- Matrix Computations-Simulation Techniques | 20 |
| III | **Statistical Modeling and Algorithms:**<br><br>Regression Models- Resampling Techniques-Bayesian Methods-Clustering and Classification-Time Series Analysis | 20 |
| IV | **Data Management and Visualization:**<br><br>Data Preprocessing-Exploratory Data Analysis (EDA)- Data Visualization-Big Data Tools-Interactive Reporting. | 15 |
| V | **Applications and Advanced Topics:**<br><br>Applications of Statistical Computing-Machine Learning Basics-Dimensionality Reduction-Statistical Computing in Research. | 15 |

**COURSE OUTCOMES:**

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | **Implement statistical computations** and algorithms using programming languages like R and Python. | K2 |
| CO2 | **Apply numerical methods** (such as root-finding, optimization, and matrix operations) to solve statistical problems. | K3 |
| CO3 | **Build and evaluate**statistical models including regression, classification, time series, and Bayesian inference. | K2 |
| CO4 | **Perform data wrangling and visualization** for real-world datasets using advanced tools and packages. | K4 |
| CO5 | **Design and execute projects** involving simulation, machine learning, and statistical computing in various domains. | K6 |

**TEXTBOOKS:**

**1.Statistical Computing with R,** by Maria L. Rizzo,2nd edition, 2021**.**

**2. Data Science and Predictive Analytics: Biomedical and Health Applications using R**, byIvo D. Dinov,2nd edition, 2023.

**REFERENCE BOOKS:**

1. Computational Statistics in Data Science,by Wiley,1st edition, 2022.
2.Handbook of Computational Statistics: Concepts and Methods, by Gentle, Hardle, Mori, 2nd edition,2021.

**WEBRESOURCE LINKS:**

1. **https://math.arizona.edu/piegorsch.html**
2. **https://link.springer.com/book/10.1007**
3. https://www.wiley.com/en-us/Computational Statistics in Data Science
4. https://link.springer.com/book/10.1007/978-3-642-21551-3

| FIRST YEAR | CORE PRACTICAL–I | COURSE CODE: P25CSCP1 |
|---|---|---|
| SEMESTER-I | ADVANCED JAVA PROGRAMMING LAB | CREDITS:3 |

**COURSE OBJECTIVES:**

- ToenablethestudentstoimplementthesimpleprogramsusingJSP,JAR
- ToprovideknowledgeonusingServlets,Applets
- TointroduceJDBCandnavigationofrecords
- TounderstandRMI&its implementation
- TointroducetoSocketprogramming

**List of Programs:**

1. Displayawelcomemessage using Servlet.
2. ToinvokeServletfromApplet.
3. Developaprogramforcalculatingthepercentageofmarksofastudentusing JSP.
4. ImplementationandExceptionHandlingconceptswithdifferenttypesofExceptions.
5. WriteaprogramtoconnectdatabaseusingJDBC.
6. Tocreateacolorpalatewithamatrixofbuttonusing Applets.
7. WriteaprograminJSPbyusingsessionobject.
8. WriteaprogramtobuildasimpleClientServerapplicationusingRMI.
9. Createanappletforacalculatorapplication.
10. Programtosendatextmessagetoanothersystemandreceivethetext message fromthe system (use socket programming).

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understand to the implement concepts of Java using HTML forms, JSP&JAR | K2 |
| CO2 | Must be capable of implementing JDBC and RMI concepts | K3 |
| CO3 | Able to write Applets with Event Handling mechanism. | K1 |
| CO4 | Create interactive web based applications using servlets and jsp | K6 |
| CO5 | Create socket based programs | K6 |

| FIRSTYEAR | COREPRACTICAL–II | COURSECODE: P25CSCP2 |
|---|---|---|
| SEMESTER-I | ADVANCED WEBTECHNOLOGIESLAB | CREDITS:3 |

**COURSEOBJECTIVES**

- LearnhowtocreatewebpagesusingHTML,CSSandJavascript.
- ImplementdynamicwebpagesusingJavascript.
- TocreatewebapplicationsusingASPandJSP.
- To design and implement different types of file structures using standard methodology.
- To create web pages with database connection.

**Implementthefollowingproblems**

1. WriteaXMLprogramforjoblistinginHTML.
2. Write a JavaScript code block, which checks the contents entered in a form's text element. If the text entered is in the lower case, convert to upper case.
3. WriteaJavaScriptcodeblock,which validates a usernameandpassword.
4. Ifeitherthenameorpasswordfieldisnotentereddisplayanerrormessage.
   *Thefieldsareentereddonotmatchwithdefaultvaluesdisplayanerror message.
   *Ifthefieldsenteredmatch,displaythewelcomemessage.
5. WriteaJavaScriptcodetodisplaythe current dateandtimeinabrowser.
6. WriteaJSPProgramforuserauthentication.
7. WriteaJSPProgramforasimpleshopping cart.
8. WriteaJSPProgramtopreparebiodataandstoreitindatabase.
9. WriteanASPProgramusingResponseandRequestObject.
10. WriteanASPProgramusingAdRotatorComponent.
11. WriteanASP Programusingdatabaseconnectivityforstudent'srecord

**COURSE OUTCOMES:**

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Design dynamic web pages using Java script | K1 |
| CO2 | CreatewebapplicationusingPHPandMySQL | K2 |
| CO3 | DevelopinteractivewebpagesusingJsp | K3 |
| CO4 | Implementing different operations on web page using ASP | K4 |
| CO5 | Ability to design and implement different types of file structures using standard methodology. | K5 |

# SEMESTER-II

| FIRST YEAR | CORE COURSE–IV | COURSECODE: P25CSCC21 |
|---|---|---|
| SEMESTER- II | ADVANCED DATABASE SYSTEMS | CREDITS: 4 |

## COURSE OBJECTIVES:

- Explore and analyze complex data models beyond the relational model, such as object-oriented, object-relational and databases.
- Gain deep insight into query evaluation techniques and query optimization strategies to improve database performance.
- Learn the architecture, design, and challenges of distributed databases, replication, and parallel query processing.
- Learn about joingtable, view and create copies of database.
- Understand the theory and implementation of transactions, including data security, integrity and Transactions.

| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | RDBMS: Structure of Relational Databases – Fundamental Relational algebra Operations – AdditionalRelational Algebra Operations – Extended Relational AlgebraOperations – Tuple RelationalCalculus –Domain Relational Calculus – EntityRelationship Model – FeaturesofGood Relational Design. | 20 |
| II | Introduction – SQL and Relational Database Management – Designing Databases:Database Design –Data Entity Relationships – The Normalization Guidelines – Reviewing of Databases – Implementing the design. | 20 |
| III | Creating and Filling the Databases- Selecting Data from Bases – Sorting Data and Other SelectionTechniques – Grouping Data and reporting it: Grouping – The Group BY clause – The Having clause –More on Nulls – Working with Multiple Tables. | 20 |
| IV | Joining Tables for Comprehensive Data Analysis – Structuring Queries with Sub Quires – CreatingUsing Classes: View towards Flexibility – Creating Views – Advantages ofViews – How View Works? –Data Modification through Views – Creating Copies of Data – Database Administrative Issues. | 15 |
| V | Security, Transaction, Performance and Integrity: Database Management in the Real world – DataSecurity – Transaction – Performance – Data Integrity – Solving Business Problems. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Demonstrate understanding of the **structure of relational databases**, including the relational model, ER diagrams, and features of good relational design . | K2 |
| CO2 | Apply **relational algebra** (basic, additional, extended) and **tuple/domain relational calculus** to express queries formally | K3 |
| CO3 | Design ER models for information needs map ER diagrams to relational schemas apply normalization principles to reduce anomalies. | K5 |
| CO4 | Design ER models for information needs; map ER diagrams to relational schemas; apply normalization principles to reduce anomalies | K4 |
| CO5 | Evaluate database schemas and enforce **data integrity** through entity/foreign keys and normalization rules | K5 |

**Text Books:**


1. " DatabaseSystemConcepts"FifthEdition,AbrahamSilberschtzHenryF. Korth, S.SudarshanMC GRAW – HILL INTERNATIONAL EDITION
2. "ThePracticalSQLHandbookUsingStructuredQueryLanguage" ThirdEdition,JudithS.Bowman Sandra L. Emerson Marcy Darnovsky.


**ReferenceBooks:**


1. "TheCompleteReferenceMySQL"TataMcGrawHillEdition,VikaramVaswani.
2. SharonAllen,Secondedition,PublishedMarch25th2005"BeginningRelationalData modeling"
3. Er.RajivChopra.,"DatabaseManagementSystems-APracticalApproach(*DBMS*)"


**Webresource Link:**

1. https://www.tutorialspoint.com/data_base/index.html
2. https://mindmajix.com/data-base

| FIRST YEAR | CORE COURSE–V | COURSECODE: P25CSCC22 |
|---|---|---|
| SEMESTER- II | PROBLEM SOLVING USING PYTHON AND R | CREDITS: 4 |

## COURSE OBJECTIVES:

- TounderstandComputationalthinkingusingPython.
- TodevelopsimplePythonprogramsforsolvingproblems.
- TomakestudentsexercisethefundamentalsofstatisticalanalysisinR environment.
- To teach the use of Python and R for data manipulation, analysis, and visualization.
- To enable students to solve real-world computational problems using appropriate programming constructs and libraries.

| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **INTRODUCTION TO PYTHON:** Introduction–Pythonoverview–Gettingstarted–Comments–Pythonidentifiers – Reserved keywords – Variables – Standard data types – Operators – StatementsandExpressions–Stringoperations–Booleanexpressions.ControlStatements: The for loop – while statement – if-else if-else statement – Input from keyboard. Functions: Introduction – Built-in functions – User defined functions – Function Definition – Function Call - Type conversion – Type coercion – Python recursive function. | **15** |
| II | **STRINGS:** Characters – String formatting operator – String formatting functions.Tuples:Tuples–Creatingtuples–Accessingvaluesintuples–Tupleassignment –Tuplesasreturnvalues–Basictupleoperations–Built-intuplefunctions.Lists:Valuesandaccessingelements–Traversingalist–Deletingelementsfrom list–Built-inlistoperators&methods.Dictionaries:Creatingdictionary– Accessing values in dictionary – Updating dictionary – Deleting elements from dictionary–Operations in dictionary-Built-in dictionary methods. | **15** |
| III | **FILES AND EXCEPTIONS:** IntroductiontoFileInputandOutput-WritingstructurestoaFile-Usingloops to process files Processing Records - Exception. Classes and Objects in Python: Overview of OOP–Data encapsulation–Polymorphism–Class definition– Creating objects – Inheritance – Multiple inheritances – Method overriding –Data encapsulation – Data hiding. | **15** |
| IV | **DATAMANIPULATION TOOLS&SOFTWARES:** Numpy: Installation- Nd array –Basic Operations -Indexing, Slicing, and Iterating-Shape Manipulation - Array Manipulation - Structured Arrays -Reading and Writing Array Data on Files. Pandas: The pandas Library: An Introduction - Installation - Introduction to pandas Data Structures - Operations between Data Structures-Function Application and Mapping - Sorting and Ranking -Correlation and Covariance - —Not a Number Data – Hierarchical Indexing and Leveling –Reading and Writing Data: CSV or Text File - HTML Files – Microsoft Excel Files. | **15** |
| V | **PROGRAMMING WITH R:** Variables-Vector,matrix,arrays–List–DataFrames–Functions–Strings– Factors – Loops – Packages –Date and Time – Files - Making packages | **15** |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| C01 | Write Python programs using Python datastructures. | K1 |
| C02 | Develop object oriented programs in Python | K6 |
| C03 | Manipulate files using Python | K3 |
| C04 | Apply the Python libraries NumPy and P and as for problem solving | K3 |
| C05 | Write R programs for data visualization. | K1 |

**Text Books:**

1. DataAnalyticsUsingPython,BhartiMotwani,Wiley,2024
2. Python: TheCompleteReference, MartinBrown,McGraw-Hill, 2022.
3. PythonProgrammingaModularApproachwithGraphics,Database,Mobile, and Web Applications – SheetalTaneja, Naveen Kumar – Pearson Publication, 2022.

**Reference books:**

1. E Balagurusamy, —Introduction to computing and problem solving using Python, McGraw Hill Publication,2024.
2. Python and R for the Modern Data Scientist: The Best of both worlds,1st edition, 2021 by O'Reilly.
3. An Introduction to R and Python for Data Analysis: A side-by-side Approach, 1steditон, 2023 by Chapman & Hall.

**Webresource link**

1. https://realpython.com/python-practice-problems/
2. https://freepdf-books.com/impractical-python-projects-playful-programming-activitiesto-make-yousmarter-book-of-2024/

| FIRST YEAR | CORE COURSE–VI | COURSECODE: P25CSCC23 |
|---|---|---|
| SEMESTER- II | MOBILE APPLICATION DEVELOPMENT | CREDITS: 4 |

## COURSE OBJECTIVES:

- AgoodunderstandingontheMobileEnvironment.
- ToKnowtheArchitectureoftheMobileApplication.
- ToUnderstandtheFeaturesofAndroid.
- To teach the design, development, testing, and deployment of mobile apps using modern programming languages and tools (e.g., Java, Kotlin, Swift, Flutter).
- To develop skills in creating user-friendly and responsive mobile interfaces with a focus on user.

| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **INTRODUCTIONTOMOBILEAPPLICATIONS:** Native and web applications - Mobile OS andDatabases. Introduction to Android: History - Features – OSS – OHA - Versions - Android devices - Setting up software – IDE. Introduction to iOS – iOS features –user interface - Using Wi-Fi – IPhone marketplace. | **15** |
| II | **ANDROIDARCHITECTURE:** Android Stack - Linux Kernel - Android Runtime - Dalvik VM – Application Framework-Android emulator- Android applications development- Virtualization – APIs – Android File system – A Basic Android Application Deployment. Android Activities: The Activity Lifecycle– Lifecycle methods– Creating Activity | **15** |
| III | **INTENTS:** Intent Filters – Activity stack. Android Services: Simple services – Binding and Querying the service – Executing services. Broadcast Receivers: Creating and managing receivers – Receiver intents. Content Providers: Creating and using content providers – Content resolver. | **15** |
| IV | **ANDROIDUI:** Android Layouts – Attributes – Layout styles - Linear – Relative – Table – Grid– Frame – Menus - Lists and Notifications - Input Controls: Buttons - Text Fields – Checkboxes-alert dialogs–Spinners-rating bar-progress bar. | **15** |
| V | **WORKINGWITHDATABASES**: SQLite coding for SQLite using Android Publishing and Internationalizing mobile applications - mobile application deployment: Game, Clock, Calendar, Convertor, Phone book. | **15** |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understandthe features and challenges of mobile devices. | K1 |
| CO2 | Knowthedifferencesbetweennativeappdevelopment,webappdevelopment and hybrid app development. | K1 |
| CO3 | ApplytheUIcomponents,multimediausage,permissions,Storageusage, read and write operations in storage. | K3 |
| CO4 | UnderstandandapplytheUIdesignforthegivenproblem. | K1 |
| CO5 | Design an application based on the user requirements. | K6 |

**Text Books:**


1. BarryBurd,"AndroidApplicationDevelopment–All-in-oneforDummies", 2ndEdition, Wiley India, 2022.
2. Lauren Darcy,ShaneConder,"SamsTeachYourselfAndroidApplicationDevelopment in 24hours", 2nd edition, Pearson Education, 2024.
3. Jerome(J.F)DiMarzio, "Android–AProgrammersGuide",McGraw Hill Education, 8th reprint, 2024.


**Reference books:**


1.DavidMark,JackNutting,JeffLaMarcheandFredericOlsson,"Beginning iOS Development: Exploring the iOS SDK",Apress, 2022.

2. Android App Development for Dummies, 3rdedition ,2023, by Michael Burton.

3.Head First Android Development: A Brain Friendly Guide ,2nd edition,2021 by David Griffiths.


**WebResourcelinks :**

1. https://www.developer.android.com
2. https://www.wired.com/2010/12/w3c-offers-a-guide-to-building-mobile-web-apps

| FIRST YEAR | CORE PRACTICAL–III | COURSE CODE: P25CSCP3 |
|---|---|---|
| SEMESTER-II | ADVANCED DATABASE SYSTEM LAB | CREDITS: 3 |

**COURSE OBJECTIVES:**

- Understand and apply fundamental SQL Data Definition Language (DDL) and Data Manipulation Language (DML) statements.
- Design normalized database schemas and enforceKeys.
- Develop procedural extensions using PL/SQL blocks, including cursors, triggers, control structures, and exception handling
- Apply Transaction Control Language (TCL) and Data Control Language (DCL)
- Generate and present data-driven reports using advanced SQL features

**LIST OF PROGRAMES:**

1. Createatable for ticket reservation details withnecessaryfields. Insert thedata values andmodify the table using INSERT, ALTER, MODIFY DROP keywords.
2. Create a table for employee details with employee number as primary key, Emp_Name,Address, designation, department, Age, Gender, DOJ, Net Salary and Gross Salary. Insertthevalues, group the employees using department and sort them.
3. Write a PL/SQL program for student table with Register No as primary key, student name,subject marks and average. Update the table by adding distinction field.
4. Create a table for railway reservation with necessary fields. Insert the data values and use DCLand TCL commands like GRAND, REVOKE, COMMIT, ROLL BACK and SAVE POINTcommands.
5. Createatableforlibrarymanagementsystemwithtwotables master andtransactiontablesand create a report based on the date of return column.
6. CreateatableforbookstoreandwriteaPL/SQLtoupdatetheratefieldwith25%incrementand display the result.
7. Updatethe above student table with result field and write a PL/SQLto split the table intoTwo based on the result field.
8. Createatableforelectricitybillcalculationusingcursorprogramming.
9. WriteaPL/SQLprogramtotriggerTicketreservationtabletodisplaytheticketbookedto CHENNAI.
10. Writeaprogramtogeneratereportforemployeedetailstable.

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | **Develop PL/SQL blocks** incorporating loops, conditional statements, and updates for domain-specific logic. | K5 |
| CO2 | **Process row-by-row operations** using cursors—for example, computing electricity bill amounts in a dedicated billing table | K1 |
| CO3 | **create PL/SQL triggers** to automate data-driven events. | K2 |
| CO4 | **combine SQL and PL/SQL techniques** to design and implement complete mini-applications. | K3 |

| | | |
|---|---|---|
| CO5 | Apply transaction control and user privileges. | K4 |

| | | |
|---|---|---|
| **FIRST YEAR** | **CORE PRACTICAL–IV** | **COURSE CODE:** |
| **SEMESTER-II** | **PROBLEM SOLVING USING PYTHON AND R LAB** | **CREDITS: 3** |

**COURSE OBJECTIVES:**

- To master the fundamentals of writing python scripts
- To create program using elementary data items
- To implement Python programs with Conditionals and loops
- To use functions for structuring Python programs
- To develop webprogramming with Django

**LIST OF PROGRAMS:**

1. Program using elementary dataitems, lists, dictionaries and tuples
2. Program using conditional branches, loops
3. Program using functions
4. Program using classes and objects
5. Program using inheritance
6. Program using polymorphism
7. Program using Numpy
8. Program using Pandas
9. Program using Matplotlib
10. Program for creating dynamic and interactive webpages using forms

*****

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Comprehendtheprogrammingskillsinpythonandwritescripts. | K1 |
| CO2 | Createpythonapplicationswithelementarydataitems, lists,dictionariesand tuples | K3 |
| CO3 | Implement the Object Oriented Programming concepts such as objects and classes, Inheritance and polymorphism | K6 |
| CO4 | Assessthe use of Python packagestoperform numerical computationsand perform data visualization | K1,K6 |
| CO5 | CreateinteractivewebapplicationsusingDjango | K1,K5 |

| FIRST YEAR | ELECTIVE COURSE–II A | COURSE CODE: P25CSE2A |
|---|---|---|
| SEMESTER-II | INTERNET OF THINGS | CREDITS: 4 |

## COURSE OBJECTIVES:

- To get familiar withtheevolutionofIOTwithitsdesignprinciples
- Tooutlinethefunctionalitiesandprotocolsofinternetcommunication
- Toanalyzethehardwareandsoftwarecomponentsneededtoconstruct IOTapplications
- ToidentifytheappropriateprotocolforAPIconstructionandwritingembeddedcode
- TorealizevariousbusinessmodelsandethicsinInternetofThings

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| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **FUNDAMENTALS OF IOT**<br>Evolution of Internet of Things – Enabling Technologies – IOT Architectures: oneM2M, IOT World Forum (IOTWF)and Alternative IOT models –Simplified IOT Architecture and Core IOT Functional Stack -–Fog, Edge and Cloud in IOT – Functional blocks of an IOT ecosystem – Sensors, Actuators, Smart Objects and Connecting Smart Objects. | 15 |
| II | **IOT PROTOCOLS**<br>IOT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and Lo Ra WAN – Network Layer: IP versions, Constrained Nodes and Constrained Networks – Optimizing IP for IOT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks – Application Transport Methods: Supervisory Control and Data Acquisition – Application Layer Protocols: CoAP and MQTT. | 15 |
| III | **DESIGNANDDEVELOPMENT**<br>Prototyping Embedded Devices: Electronics - Embedded Computing Basics – Arduino - Raspberry Pi - Beagle Bone Black - Electric Imp. Prototyping the Physical Design: Non digital Methods - Laser Cutting - 3D printing - CNC Milling - Repurposing/Recycling. | 15 |
| IV | **PROTOTYPING ONLINE COMPONENTS**<br>Getting started with an API - Writing a New API - Real-Time Reactions - Other Protocols. Techniques for Writing Embedded Code: Memory Management - Performance and Battery Life – Libraries - Debugging. | 15 |
| V | **BUSINESS MODELS**<br>History of Business Models – Model – Internet of Starting up – Lean Startups. Moving to Manufacture: Designing Kits - Designing Printed circuit boards – Certification – Costs - Scaling Up Software. Ethics: Privacy – Control – Environment – Solutions. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Comprehend the IOT evolution with its architecture and sensors | K5 |
| CO2 | Understand the networking concepts for communication and underlying IOT Protocols | K2 |
| CO3 | Assess the embedded technologies and develop prototypes for the IOT Products | K6 |
| CO4 | Evaluate the use of Application Programming Interface and design an API for IoT in real time | K5 |
| CO5 | Recognize the ethics of business models and perform security analysis | K1 |

**Text Books:**

1. David Hanes, Gonzalo Salgueiro, Patrick Grossetete,RobBartonand Jerome Henry, —IoTFundamentals: Networking Technologies, Protocols and Use Cases for Internet ofThings, Cisco Press, 2020 (UNIT I and II)
2. Adrian McEwen and Hakim Cassimally, "Designing the Internet of Things", Wiley, 2019. (UNIT III, IV and V)

**Reference Books:**

1. OvidiuVermesanandPeterFriess, "Internet ofThings–FromResearchandInnovationtoMarket Deployment", River Publishers, 2020.
2. PeterWaher,"LearningInternetofThings", Packet Publishing, 2019.
3. Donald Norris, "The Internet of Things: Do-It-Yourself atHome Projects for Arduino, Raspberry Pi and BeagleBoneBlack",McGraw Hill, 2020.

**Webresource link:**

1. https://www.ciscopress.com/store/iot-fundamentals-networking-technologies
2. https://www.oreilly.com/library/view/iot-fundamentals-networking
3. https://www.wiley.com/en-us/Fundamentals of Internet of Things

| FIRST YEAR | ELECTIVE COURSE–II B | COURSE CODE: P25CSE2B |
|---|---|---|
| SEMESTER-II | ARTIFICIAL INTELLIGENCE | CREDITS: 4 |

## COURSE OBJECTIVES:

- ToAcquireKnowledgeonvariousAITechniquesandExpertSystems
- To Richeknowledgeregardingheuristicsearch,Knowledgerepresentationand Expert systems
- The course aims to introduce key AI concepts such as machine learning, reasoning, natural language processing, robotics,
- To preparing students to apply AI technologies to real-world problems effectively and ethically.
- To represent knowledge of problem-solving techniques

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| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **Introduction**: AI Problems–AItechniques–Criteriaforsuccess.Problems,Problem Spaces. Search: State space search – Production Systems – Problem Characteristics – Issues in design of Search. | **15** |
| II | **Heuristic Search techniques:**Generate and Test Hill Climbing Best Fist,Problemreduction,ConstraintSatisfaction,Means-endanalysis. | **15** |
| III | **Knowledge representation issues:**Representationsand mappings –Approachesto Knowledge representations – Issues in Knowledge representations – Frame Problem. | **15** |
| IV | **Using Predicate Logic:** Representing simple facts in logic – Representing Instance and Isa relationships – Computable functions and predicates – Resolution – Natural deduction. | **15** |
| V | **Representing knowledgeusing rules:** Procedural Vs Declarative knowledge – Logic programming – Forward Vs Backward reasoning – Matching – Control knowledgeBrief explanation of Expert Systems- Definition- Characteristics-architecture- Knowledge Engineering- Expert System Life Cycle-Knowledge Acquisition Strategies- Expert System Tools. | **15** |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Explain fundamental AI concepts, including AI problems, techniques, and criteria for successful AI systems. | K5 |
| CO2 | Implement heuristic search strategies including generate-and-test, hill climbing, best-first search, problem reduction, constraint satisfaction, and means-end analysis. | K5 |
| CO3 | Describe and apply various search techniques such as state space search, production systems, and heuristic methods to solve AI problems. | K2 |
| CO4 | Evaluate problem characteristics and design considerations critical to efficient and effective search algorithms. | K5 |
| CO5 | Define problem spaces and analyze the characteristics and components of AI problems. | K1 |

**Text Books:**

1.ElaineRich andKevinKnight,ShivaShankarNair,"Artificial Intelligence",McGraw-Hill Companies, 3rd edition,2021.

2.Aritifical Intelligence: A modern Approach ,4th edition ,2020 by Stuart J.Russell&Peter Norvig.

3.Aritificial Intelligence: A Text book, First edition ,2021 by CharuC. Aggarwal.

**ReferenceBooks:**

1. StuartRussell&PeterNorvig, "ArtificialIntelligence a ModernApproach", Perason,2nd Edition.
2. GeorgeFLuger,"ArtificialIntelligence",Pearson2002,4thEdition.
3. VSJanakiRaman,K . Sarukesi,P. Gopala Krishnan,"FoundationsofArtificialIntelligentand Expert Systems", MacMillan India limited.

**WebResources:**

1. NPTEL&MOOC courses titled Artificial Intelligence and Expert Systems
2. https://nptel.ac.in/courses
3. https://indiaai.gov.in/article/five-free-online-books-on-artificial intelligence

| FIRST YEAR | NON MAJOR ELECTIVE COURSE–IA | COURSE CODE: P25CSNM1 |
|---|---|---|
| SEMESTER-II | CRYPTOGRAPHY | CREDITS: 2 |

## COURSE OBJECTIVES:

- Understand the OSI security architecture, including types of security attacks, mechanisms, and services that ensure network security.
- Explain classical encryption techniques such as symmetric ciphers, substitution ciphers (Caesar, monoalphabetic, Playfair, polyalphabetic), transposition ciphers, and steganography methods.
- Describe the principles of block ciphers and the Data Encryption Standard (DES), including the strengths and weaknesses of DES.
- Understand and apply the RSA algorithm for public-key cryptography and analyze its security implications.

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| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **Introduction:** TheOSI security Architecture –Security Attacks–Security Mechanisms –Security Services– A model for network Security. | 10 |
| II | **Classical Encryption Techniques:** Symmetric cipher model – Substitution Techniques: Caesar Cipher – Monoalphabetic cipher – Play fair cipher – Poly Alphabetic Cipher – Transposition techniques – Stenography. | 10 |
| III | **BlockCipherandDES:**Block cipher Principles–DES –TheStrengthofDES – RSA:TheRSAalgorithm. | 9 |
| IV | **NetworkSecurityPractices:**IPSecurityoverview–IPSecurityarchitecture – AuthenticationHeader. Web Security: SecureSocketLayer and Transport Layer Security – Secure Electronic Transaction. | 8 |
| V | Intruders–Malicious software –Firewalls. | 8 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO 1 | Analyzethevulnerabilitiesinanycomputingsystemandhencebeabletodesignasecuritysolution. | K4 |
| CO 2 | Applythedifferentcryptographicoperationsofsymmetriccryptographic algorithms. | K3 |
| CO 3 | Applythedifferentcryptographicoperationsofpublickeycryptography. | K3 |
| CO 4 | ApplythevariousAuthenticationschemestosimulatedifferentapplications. | K3 |
| CO 5 | Analyzethevulnerabilitiesinanycomputingsystemandhencebeabletodesignasecuritysolution. | K4 |

**Text Books:**

1. William Stallings, "CryptographyandNetworkSecurityPrinciplesandPractices", 8th edition,2024.
2. Cryptography and Network Security, 4thedition ,2020, by AtulKahate.
3. Understanding Cryptography: from Established Symmetric and Asymmetric Ciphers to post Quantum Algorithm ,2nd edition, 2024 by Christof Paar& Jan Pelzl.

**ReferenceBooks:**

1. BehrouzA. Foruzan,"CryptographyandNetworkSecurity",TataMcGraw-Hill, 2022.
2. Cryptography and Network Security, 1steditон ,2020 by RathnakarAchary.

**Webresource link:**

1. https://en.wikipedia.org/wiki/Cryptography
2. https://cryptopals.com/
3. https://www.khanacademy.org/computing/computer-science/cryptography

| FIRST YEAR | NON MAJOR ELECTIVE COURSE–IB | COURSE CODE: P25CSNM1 |
|---|---|---|
| SEMESTER-II | FUNDAMETALS OF BIGDATA | CREDITS: 2 |

## COURSE OBJECTIVES:

- UnderstandtheBigDataPlatformanditsUsecases
- ProvideanoverviewofApacheHadoop
- ProvidehandsonHadoop Eco System
- To introduce the core concepts, characteristics, and significance of Big Data in today's world.
- To explain the architecture, technologies, and tools used in Big Data processing and management.

| UNITS | CONTENTS | HOURS |
|---|---|---|
| I | **Introduction to BigDataand Hadoop**<br>Types of Digital Data, Introduction to Big Data, Big DataAnalytics, History of Hadoop, Apache Hadoop, Analyzing Data with Unix tools, Analyzing Data with Hadoop, Hadoop Streaming, Hadoop Echo System, IBM Big Data Strategy. | **10** |
| II | **HDFS (Hadoop Distributed File System)**<br>The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, | **10** |
| III | **MapReduce**<br>Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features. | **9** |
| IV | **Hadoop EcoSystem**<br>Introduction to Pig-Hive-H Base-Big SQL | **8** |
| V | **DataAnalytics with R**<br>MachineLearning:Introduction, Supervised Learning,UnsupervisedLearning,Collaborative Filtering | **8** |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | IdentifyBigDataanditsBusinessImplications. | K1 |
| CO2 | ListthecomponentsofHadoopandHadoopEco-System. | K1 |
| CO3 | AccessandProcessDataonDistributedFileSystem. | K5 |
| CO4 | ManageJobExecutioninHadoopEnvironment. | K5 |
| CO5 | Understanding the machine learning concepts. | K2 |

**Text Books:**


1. Tom White "Hadoop: The Definitive Guide"Third Edit on, O'reilyMedia,2021.
2. Seem an Acharya,SubhasiniChellappan,"Big Data Analytics"Wiley2022.
3. The Big Data Text book, 2nd edition, 2024 by Ghislainfourny.


**Reference Books:**


1. Michael Berthold, David.Hand," Intelligent Data Analysis", Springer,2020.
2. Fundamentals of BigData Analytics, 1st edition ,2022 by Sylvia Grace,2021.


**Webresource link:**

1. https://github.com/bsharvey/bsharvey.github.io/blob/master/assets/books/bigdatafundamentals
2. https://www.oreilly.com/library/view/big-data-fundamentals

# SEMESTER-III

| SECOND YEAR | CORECOURSE–VII | COURSECODE: P25CSCC31 |
|---|---|---|
| SEMESTER- III | BIG DATA ANALYTICS | CREDITS: 4 |

## COURSE OBJECTIVES:

- Understand the fundamentals of Big Data, its characteristics, challenges, and differences from traditional data processing systems.
- Analyse modern Big Data technologies including Hadoop, NoSQL, and distributed computing frameworks.
- Develop skills in handling Big Data using MongoDB and implementing MapReduce programming models.
- Apply data processing techniques using Hive and Pig for large-scale data analysis and querying.
- Gain practical knowledge of Big Data analytics tools, data science concepts, and real-world applications.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **Big Data and Analytics:** Classification of Digital Data: Structured Data- Semi Structured Data and Unstructured Data. Introduction to Big Data: Characteristics – Evolution – Definition - Challenges with Big Data - Other Characteristics of Data - Big Data - Traditional Business Intelligence versus Big Data - Data Warehouse and Hadoop. | 20 |
| II | **Technology Landscape:** NoSQL, Comparison of SQL and NoSQL, Hadoop -RDBMS Versus Hadoop - Distributed Computing Challenges – Hadoop Overview - Hadoop Distributed File System - Processing Data with Hadoop - Managing Resources and Applications with Hadoop YARN - Interacting with Hadoop Ecosystem. | 20 |
| III | **Mongo dB and Map reduce Programming:** MongoDB: Mongo DB - Terms used in RDBMS and Mongo DB - Data Types - MongoDB Query Language. MapReduce:Mapper –Reducer –Combiner–Practitioner–Searching –Sorting–Compression. | 20 |
| IV | **Hive:** Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing– Views - Sub- Query– Joins – Aggregations – Group by and Having– RC File - Implementation- Hive User Defined Function - Serialization and De serialization. | 15 |
| V | **Pig:** Introduction- Anatomy– Features–Philosophy- Use Case for Pig- Pig Latin Overview- Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - Relational Operators - Eval Function - Complex Data Types - Piggy Bank - User-Defined Functions - Parameter Substitution – Diagnostic Operator - Word Count Example using Pig - Pig at Yahoo! - Pig Versus Hive. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand Big Data concepts including data types, characteristics, challenges, and analytics fundamentals. | **K2** |
| **C02** | Analyse Big Data technologies such as Hadoop, NoSQL, and distributed computing frameworks. | **K4** |
| **C03** | Apply MongoDB concepts and implement Map Reduce programming for data processing. | **K3** |
| **C04** | Develop data analysis solutions using Hive for querying and managing large datasets. | **K6** |
| **C05** | Implement data processing and analytics using Pig and compare it with Hive for real-time applications. | **K6** |

**Text Books:**


1. Tom White, "Hadoop: The Definitive Guide", O'Reilly Publications, Fourth Edition, 2021. https://www.oreilly.com/library/view/hadoop-the-definitive/9781491901687/
2. Kristina Chodorow, "MongoDB: The Definitive Guide", O'Reilly Publications, Third Edition,2020. https://www.oreilly.com/library/view/mongodb-the-definitive/9781491954454/
3. Foster Provost, Tom Fawcett, "Data Science for Business", O'Reilly Publications, First Edition (Reprint), 2021. https://www.oreilly.com/library/view/data-science-for/9781449374273/


**Reference books:**


1. Balamurugan Balusamy, Nandhini Abirami R, Seifedine Kadry, Amir H. Gandomi, "Big Data: Concepts, Technology, and Architecture", Wiley Publications, First Edition, 2021. https://books.google.com/books/about/Big_Data.html?id=u7IfEAAAQBAJ
2. Nitin Kumar, "Big Data Using Hadoop and Hive", Mercury Learning & Information, First Edition, 2021. https://books.google.com/books/about/Big_Data_Using_Hadoop_and_Hive.html?id=IUlgzgEACAAJ
3. S. Chandramouli et al., "Big Data Analytics", Universities Press, First Edition, 2024. https://www.universitiespress.com/details?id=9789393330468
4. Matt Mueyon, "Data Engineering with Apache Hadoop and Hive", Amazon KDP, First Edition, 2024. https://books.google.com/books/about/Data_Engineering_with_Apache_Hadoop_and.html?id=2qSr0AEACAAJ
5. Simhadri Govindappa, "Ultimate Big Data Analytics with Apache Hadoop: Master Big Data Analytics using Spark, Hive and Python", Orange Education, First Edition, 2024. https://play.google.com/store/books/details?id=RyNIEQAAQBAJ

| SECONDYEAR | CORECOURSE–VIII | COURSECODE: P25CSCC32 |
|---|---|---|
| SEMESTER-III | COMPILER DESIGN | CREDITS:4 |

## COURSE OBJECTIVES:

- Understand the fundamental concepts of compilers, phases, and error handling mechanisms.
- Analyse lexical analysis techniques using regular expressions and finite automata.
- Apply parsing techniques for syntax analysis including top-down and bottom-up approaches.
- Develop intermediate code generation and syntax-directed translation methods.
- Evaluate code optimization techniques and implement efficient target code generation.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **INTRODUCTIONTOCOMPILERS:** Compilers – Analysis – Synthesis model of compilation – Analysis of the source program – The phases of a compiler –Cousins of the compiler construction tools – Error handling. | 20 |
| II | **LEXICALANALYZER:** Lexical analysis – Role of lexical analyzer – Tokens, Patterns and lexemes – Input buffering – Specification of tokens – Regular expressions – Recognition of tokens – Transition diagrams – Implementing a transition diagram – Finite Automata- Regular expression to NFA – Conversion of NFA to DFA | 20 |
| III | **SYNTAXANALYZER:** Syntax analysis – Role of parser – Context-free grammars – Derivations – Writing a grammar – Top Down parsing – Recursive descent parsing – Predictive parsers – Non- recursive predictive parsers – Construction of predictive parsing tables- Bottom up parsing- Handles-Shift reduce parser Operator precedence parsing -LR parsers–Canonical collection of LR(0)items –Constructing SLR parsing tables. | 20 |
| IV | **INTERMEDIATECODEGENERATION:** code Representation-Three generation-Intermediate Address Code-Assignment languages-Graphical statements-Boolean expressions – Flow of Control Statements – Case Statements - Syntax directed translation of case statements | 15 |
| V | **CODEOPTIMIZATIONANDCODEGENERATION:** An Organization for an Optimizing Compiler-the Principle sources of optimization- Function Preserving Transformations-Common Sub expression Copy propagation- Optimization of basic blocks-The use of Algebraic identities Loops in flow graphs- Code generation- issues in the design of a code generator The target machine. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Explain the structure of a compiler, phases of compilation, and the analysis–synthesis model used in compiler design. | **K2** |
| **C02** | Design and implement lexical analyzers using regular expressions, finite automata, NFA and DFA conversion techniques. | **K4** |
| **C03** | Analyze and construct parsers using top-down and bottom-up parsing techniques for context-free grammars. | **K3** |
| **C04** | Generate intermediate code representations and apply syntax-directed translation methods for expressions and control statements. | **K6** |
| **C05** | Analyze compiler optimization techniques and design efficient code generation strategies for target machine architectures. | **K6** |

## Text Book

- Compilers: Principles, Techniques, and Tools — Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Pearson Education,2024.

## Reference Books


1. Engineering a Compiler — Keith Cooper & Linda Torczon, Elsevier, 2022.
2. Modern Compiler Implementation in C — Andrew W. Appel, Cambridge University Press.
3. Introduction to Compiler Design — Torben Ægidius Mogensen, Springer, 2021.

| SECOND YEAR | CORE COURSE–IX | COURSE CODE: P25CSCC33 |
|---|---|---|
| SEMESTER- III | CRYPTOGRAPHY AND NETWORK SECURITY | CREDITS: 4 |

## COURSE OBJECTIVES:

- Understand fundamental concepts of network security, attacks, services, and security models.
- Analyse classical and modern cryptographic techniques including symmetric and public key cryptography.
- Apply authentication mechanisms, digital signatures, and secure communication protocols.
- Evaluate network security implementations such as IP security, SSL/TLS, and email security systems.
- Assess system-level security including intrusion detection, malware protection, and firewall design.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **INTRODUCTION:** Security Trends - The OSI Security Architecture - Security Attacks - Security Services - A model for network Security. Classical Encryption Techniques: Symmetric Cipher Model - Substitution Techniques -Transposition Techniques – Stenography. | 20 |
| II | **BLOCK CIPHERS AND THE DES:** Block cipher Principles - The DES - The Strength of DES - Differential and Linear Crypt Analysis. Public key cryptography and RSA: Principles of Public Key Cryptosystems – The RSA Algorithm. | 20 |
| III | **DIGITAL SIGNATURES AND AUTHENTICATION PROTOCOLS:** Digital Signatures - Authentication Protocols - Digital Signature Standard. Authentication Applications: Kerberos - X.509 Authentication Service, Public-Key Infrastructure. Email Security: Pretty Good Privacy - S/MIME. | 20 |
| IV | **IP SECURITY:** IP Security Overview- IP Security Architecture-Authentication Header – Encapsulating Security Payload. Web Security: Security Considerations - SSL and TLS-SET. | 15 |
| V | **SYSTEM SECURITY:** Intruders-Intrusion Detection Password Management. Malicious Software: Viruses and Related Threats. Firewalls: Design Principles - Trusted systems. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand security architecture, attacks, and services in network environments. | **K2** |
| **C02** | Apply cryptographic techniques including DES and RSA algorithms. | **K3** |
| **C03** | Analyse digital signatures, authentication protocols, and security applications. | **K4** |
| **C04** | Evaluate network security protocols such as IPsec, SSL/TLS, and email security. | **K5** |
| **C05** | Assess system security mechanisms including intrusion detection and firewall design. | **K5** |

**Text Books:**

1. Cryptography and Network Security Principles and Practice, William Stallings, Pearson Publications, Seventh Edition (Reprint), 2021.https://www.pearson.com
2. Network Security Essentials Applications and Standards, William Stallings, Pearson Publications, Sixth Edition (Reprint), 2020.https://www.pearson.com
3. Computer Security Principles and Practice, Lawrie Brown, Pearson Publications, Fourth Edition (Reprint), 2021. https://www.pearson.com

**Reference books:**

1. Introduction to Modern Cryptography, Jonathan Katz, Yehuda Lindell, CRC Press, Third Edition (Reprint), 2024.https://www.routledge.com
2. Applied Cryptography Protocols Algorithms and Source Code in C, Bruce Schneider, Wiley Publications, Reprint Edition, 2024.https://www.wiley.com
3. Network Security Private Communication in a Public World, Charlie Kaufman, Radia Perlman, Mike Speciner, Pearson Publications, Second Edition (Reprint), 2024. https://www.pearson.com
4. Security Engineering A Guide to Building Dependable Distributed Systems, Ross Anderson, Wiley Publications, Third Edition (Reprint), 2022. https://www.wiley.com
5. Cryptography Engineering Design Principles and Practical Applications, Niels Ferguson, Tadayoshi Kohno, Wiley Publications, Reprint Edition, 2023. https://www.wiley.com

| SECOND YEAR | ELECTIVE COURSE–III A | COURSE CODE: P25CCSE3A |
|---|---|---|
| SEMESTER-III | MULTICORE COMPUTER ARCHITECTURE | CREDITS:4 |

**COURSE OBJECTIVES:**

- Tounderstandthemicro-architecturaldesignofprocessors
- To learn about the various techniques used to obtain performanceimprovement and power savings in current processors
- TogainknowledgeindistributedandParallelComputingArchitectureSyllabus

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | Review of Fundamentals of CPU, Memory and IO–Trends in Technology, Power, Energy and Cost, Dependability – Performance Evaluation. | 20 |
| II | ILP Concepts – Pipelining Overview – Compiler Techniques for Exposing ILP – Dynamic Branch Prediction– Dynamic Scheduling – Multiple Instruction Issue – Hardware Based Speculation – Static Scheduling – Multi-Threading – Limitations of ILP – Case Studies. | 20 |
| III | Vector Architecture –SIMD Extensions–Graphics Processing Units–Loop Level Parallelism. | 20 |
| IV | Symmetric and Distributed Shared Memory Architectures – Performance Issues – Synchronization – Models of Memory Consistency – Case Studies: Intel I7 Processor, SMT & CMP Processors. | 15 |
| V | Cache Performance – Reducing Cache Miss Penalty and Miss Rate – Reducing Hit Time – Main Memory and Performance – Memory Technology. Types of Storage Devices– Buses– RAID – Reliability, Availability and Dependability – I/O Performance Measures. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Explain the fundamentals of computer architecture including CPU, memory, I/O systems, and analyze trends in performance, power, energy, cost, and dependability. | **K2** |
| **C02** | Analyze Instruction Level Parallelism (ILP) techniques such as pipelining, dynamic scheduling, branch prediction, hardware speculation, and multithreading to improve processor performance. | **K3** |
| **C03** | Evaluate parallel processing architectures including vector processors, SIMD architectures, GPUs, and loop-level parallelism techniques. | **K4** |
| **C04** | Describe shared memory multiprocessor architectures, synchronization mechanisms, memory consistency models, and assess the performance of multicore and SMT processors. | **K5** |
| **C05** | Analyze cache and memory hierarchy performance, storage systems, RAID architectures, buses, and I/O performance metrics for efficient computer system design. | **K5** |

**TEXTBOOK**

1. JohnLHennessey,DavidAPatterson,"ComputerArchitectureAQuantitativeApproach" Morgan Kaufmann Elsevier ,Fifth Edition 2021.
2. Computer Architecture: A Quantitative Approach By John L. Hennessy and David A. Patterson Morgan Kaufmann Publishers, 6th Edition 2024.
3. Computer Organization and Design By David A. Patterson and John L. Hennessy Elsevier / Morgan Kaufmann, 5th Edition 2024

**REFERENCEBOOKS:**

1. KaiHwangandFayeA.Briggs,"Computer Architecture and Parallel Processing", McGraw-HillInternational Editions,2021.
2. SimaD,FountainT,KacsukP,"AdvancedComputerArchitectures:ADesign Space Approach", Addison Wesley., 2025.

**WEBSOURCE LINKS:**

- Coursera – Computer Architecture Course
- NPTEL Computer Architecture Courses
- Stanford Computer Architecture Resources

| SECOND YEAR | ELECTIVE COURSE–III B | COURSE CODE: P25CCSE3B |
|---|---|---|
| SEMESTER-III | AGILE TECHNOLOGIES | CREDITS:4 |

## COURSE OBJECTIVES:

- Understand Agile principles, methodologies, and differences from traditional software development models.
- Analyse Agile processes such as Scrum, XP, Lean, and Feature Driven Development.
- Apply knowledge management techniques in Agile environments for effective collaboration.
- Evaluate Agile approaches in requirements engineering and handling dynamic requirements.
- Assess Agile quality assurance practices including testing, metrics, and global software development.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **INTRODUCTION:** Theories for Agile Management – Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agility in Design, Testing – Agile Documentations – Agile Drivers, Capabilities and Values | 20 |
| II | **AGILE PROCESS:** Lean Production - SCRUM, Crystal, Feature Driven Development- Adaptive Software Development - Extreme Programming: Method Overview – Lifecycle – Work Products, Roles and Practices. | 20 |
| III | **AGILITY AND KNOWLEDGE MANAGEMENT**: Agile Information Systems – Agile Decision Making - Earl_S Schools of KM – Institutional Knowledge Evolution Cycle – Development, Acquisition, Refinement, Distribution, Deployment , Leveraging – KM in Software Engineering – Managing Software Knowledge – Challenges of Migrating to Agile Methodologies – Agile Knowledge Sharing – Role of Story-Cards – Story-Card Maturity Model (SMM). | 20 |
| IV | **AGILITY AND REQUIREMENTS ENGINEERING:** Impact of Agile Processes in RE–Current Agile Practices – Variance – Overview of RE Using Agile – Managing Unstable Requirements – Requirements Elicitation – Agile Requirements Abstraction Model – Requirements Management in Agile Environment, Agile Requirements Prioritization – Agile Requirements Modeling and Generation – Concurrency in Agile Requirements Generation. | 15 |
| V | **AGILITY AND QUALITY ASSURANCE:** Agile Product Development – Agile Metrics – Feature Driven Development (FDD) – Financial and Production Metrics in FDD – Agile Approach to Quality Assurance - Test Driven Development – Agile Approach in Global Software Development. | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand Agile principles, values, and various Agile methodologies. | **K2** |
| **C02** | Analyse Agile development processes such as Scrum, XP, and Lean models. | **K4** |
| **C03** | Apply knowledge management and decision-making techniques in Agile environments. | **K3** |
| **C04** | Develop and manage Agile requirements using modern engineering practices. | **K6** |
| **C05** | Evaluate Agile quality assurance techniques including testing and metrics. | **K5** |

**Text Books:**


1. Agile Software Development Principles Patterns and Practices, Robert C. Martin, Pearson Publications, First Edition (Reprint), 2020. https://www.pearson.com
2. Agile Project Management with Scrum, Ken Schwaber, Microsoft Press, First Edition (Reprint), 2021. https://www.microsoftpressstore.com
3. Extreme Programming Explained Embrace Change, Kent Beck, Addison-Wesley Publications, Second Edition (Reprint), 2020.https://www.pearson.com


**Reference books:**


1. Agile Estimating and Planning, Mike Cohn, Pearson Publications, Reprint Edition, 2020. https://www.pearson.com
2. User Stories Applied For Agile Software Development, Mike Cohn, Addison-Wesley Publications, Reprint Edition, 2021.https://www.pearson.com
3. Lean Software Development An Agile Toolkit, Mary Poppendieck, Addison-Wesley Publications, Reprint Edition, 2021.https://www.pearson.com
4. The Scrum Guide, Jeff Sutherland, Scrum.org, Updated Version, 2020. https://scrumguides.org
5. Agile Testing A Practical Guide for Testers and Agile Teams, Lisa Crispin, Janet Gregory, Addison-Wesley Publications, Reprint Edition, 2020.https://www.pearson.com

| SECONDYEAR | COREPRACTICAL–V | COURSE CODE: P25CSCP5 |
|---|---|---|
| SEMESTER-III | BIG DATA ANALYTICS LAB | CREDITS:3 |

**COURSE OBJECTIVES:**

- Understand practical implementation of Big Data tools and distributed computing concepts.
- Apply Hadoop ecosystem components for storing and processing large datasets.
- Develop skills in NoSQL databases and MongoDB query processing.
- Implement data analysis using Hive and Pig scripting techniques.
- Analyse real-time datasets using Big Data frameworks and tools.

**LIST OF PROGRAMS:**

1. Display basic file operations in HDFS using Hadoop commands.
2. To implement Word Count application using MapReduce.
3. Develop a program to perform sorting and searching using MapReduce.
4. Implementation of MongoDB operations with CRUD and exception handling.
5. Write a program to connect and manipulate data using MongoDB Query Language.
6. To create and execute Hive queries for table creation and data manipulation.
7. Write a program in Hive using partitioning, bucketing, and aggregation functions.
8. Write a program to build data processing scripts using Apache Pig.
9. Create a Pig program for word count and relational operations on datasets.
10. Program to compare data processing using Hive and Pig for a given dataset.

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understand Big Data tools and environment setup including Hadoop ecosystem. | K2 |
| CO2 | Apply HDFS commands and manage distributed data storage. | K3 |
| CO3 | Implement MongoDB operations and MapReduce programming. | K3 |
| CO4 | Develop data querying and analysis using Hive. | K6 |
| CO5 | Apply Pig scripting for large-scale data processing and analytics. | K6 |

| SECONDYEAR | COREPRACTICAL–VI | COURSECODE: P25CSCP6 |
|---|---|---|
| SEMESTER-III | CRYPTOGRAPHY AND NETWORK SECURITY LAB | CREDITS:3 |

## COURSE OBJECTIVES:

- Understand fundamental cryptographic techniques through implementation of classical and modern encryption algorithms.
- Apply symmetric and substitution-based encryption methods such as Caesar, Hill, and DES algorithms.
- Analyse network security concepts including port security, packet capturing, and traffic analysis.
- Develop practical skills in implementing digital signatures and authentication mechanisms.
- Evaluate system and network vulnerabilities through controlled security experiments such as malware analysis and attack simulations.

## LIST OF PROGRAMS:

1. Write a program that contains a string (char pointer) with a value 'Hello world'. The program should XOR each character in the string with 0 and display the result.
2. Write a program to perform encryption and decryption using the Ceaser Cipher.
3. Write a program to perform encryption and decryption using the Hill Cipher.
4. Write a program to perform encryption and decryption using the Substitution Cipher.
5. Write a program to perform encryption and decryption using the DES algorithm.
6. Connect to switch with a computer and enable the port security.
7. Defeating malware using Building Trojans and Rootkit hunter.
8. Implement signature scheme–Digital Signature Standard.
9. Identifyandcapturetheusernameandpasswordinasamenetworkusingwireshark
10. Implement Man-in-the-middle attack and Session hijacking.

## COURSE OUTCOMES:

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understand basic cryptographic operations and simple encryption techniques. | K2 |
| CO2 | Apply classical encryption algorithms such as Caesar, Hill, and substitution ciphers. | K3 |
| CO3 | Implement modern cryptographic methods including DES and digital signature schemes. | K3 |
| CO4 | Analyse network traffic and identify security threats using tools like Wireshark. | K4 |
| CO5 | Evaluate system and network vulnerabilities through security mechanisms and attack simulations. | K5 |

| SECOND YEAR | NON MAJOR ELECTIVE COURSE–II A | COURSE CODE: P25CSNM2A |
|---|---|---|
| SEMESTER- III | DHTML AND XML | CREDITS: 2 |

**COURSE OBJECTIVES:**

- Understand the fundamentals of web page design using HTML and basic web technologies.
- Apply HTML elements, forms, and multimedia integration for interactive web pages.
- Analyse styling techniques using CSS and structure data using XML.
- Develop dynamic web pages using DHTML and Document Object Model (DOM).
- Implement client-side scripting using JavaScript for interactive web applications.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **HTML:** HTML-Introduction-tag basics- page structure-adding comments working with texts, paragraphs and line break. Emphasizing test- heading and horizontal rules-list-font size, face and color-alignment-links-tables- frames. | 10 |
| II | **Forms & Images Using HTML:** Graphics: Introduction-How to work efficiently with images in web pages, image maps, GIF animation, adding multimedia, data collection with html forms textbox, password, list box, combo box, text area, tools for building web page front page. | 10 |
| III | **XML & DHTML:** Cascading style sheet (CSS)-what is CSS-Why we use CSS-adding CSS to your web pages- Grouping styles-extensible markup language (XML). | 9 |
| IV | **Dynamic HTML:** Document object model(DCOM)-Accessing HTML & CSS through DCOM Dynamic content styles & positioning-Event bubbling-data binding. | 8 |
| V | **Java Script:** Client side scripting, What is JavaScript, How to develop JavaScript, simple JavaScript, variables, functions, conditions, loops and repetition. | 8 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand HTML structure, tags, and basic web page design concepts. | **K2** |
| **C02** | Apply HTML forms, images, and multimedia elements in web development. | **K3** |
| **C03** | Analyse and implement CSS and XML for styling and data representation. | **K4** |
| **C04** | Develop dynamic web pages using DHTML and DOM manipulation techniques. | **K6** |
| **C05** | Implement JavaScript for client-side validation and interactive features. | **K6** |

**Text Books:**


1. Web Programming Building Internet Applications, Chris Bates, Wiley Publications, Third Edition (Reprint), 2021.https://www.wiley.com
2. HTML and CSS Design and Build Websites, Jon Duckett, Wiley Publications, Reprint Edition, 2020.https://www.wiley.com
3. JavaScript and JQuery Interactive Front-End Web Development, Jon Duckett, Wiley Publications, Reprint Edition, 2021.https://www.wiley.com


**Referencebooks:**


1. Learning Web Design, Jennifer Niederst Robbins, O'Reilly Publications, Fifth Edition (Reprint), 2021.https://www.oreilly.com
2. Web Technologies Black Book, Kogent Learning Solutions, Dreamtech Press, Reprint Edition, 2020.https://www.dreamtechpress.com
3. Beginning JavaScript, Jeremy McPeak, Wiley Publications, Reprint Edition, 2020. https://www.wiley.com
4. XML in a Nutshell, Elliotte Rusty Harold, O'Reilly Publications, Reprint Edition, 2021. https://www.oreilly.com
5. JavaScript The Good Parts, Douglas Crockford, O'Reilly Publications, Reprint Edition, 2020.https://www.oreilly.com

| SECOND YEAR | NON MAJOR ELECTIVE COURSE–II B | COURSE CODE: P25CSNM2B |
|---|---|---|
| SEMESTER- III | CYBER LAW AND CRIMES | CREDITS: 2 |

## COURSE OBJECTIVES:

- To introduce the fundamentals of the Internet, World Wide Web, cyber law, and information technology concepts.
- To provide knowledge about the evolution of cyber laws in India and the provisions of the Information Technology Act 2000.
- To understand the societal, ethical, legal, and security issues related to Information Technology and cyberspace.
- To familiarize students with different types of cyber crimes, cyber frauds, investigation procedures, and preventive measures.
- To develop awareness about privacy, intellectual property rights, security, and career opportunities in Information Technology.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | Introduction-History of Internet and World Wide Web-Need for cyber law-Cyber crime on the rise-Important terms related to cyber law | 10 |
| II | Cyber law in India - Need for cyber law in India - History of cyber law in India- Information Technology Act, 2000 - Overview of other laws amended by the IT Act, 2000 - National Policy on Information Technology 2012 | 10 |
| III | Information Technology – why we need IT- work of IT companies – advantages and disadvantages of IT- Overview of the Information Technology Act | 9 |
| IV | Societal Impacts of Information Technology: Introduction, Privacy, Security and Integrity of Information, Disaster Recovery, Intellectual Property Rights, Careers in Information Technology. | 8 |
| V | Cyber crimes and Cyber Frauds - Definition of cyber crime - First Cyber crime - Types of cyber frauds - Cyber frauds in India - Preventive measures -. Cyber crimes- Who commits cyber crimes - Penalties and offences under the IT Act,2000-Offencesunder other legislations-Investigation of cyber crimes in India | 8 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **CO1** | Explain the fundamentals of the Internet, World Wide Web, cyber law, and the need for cyber regulations in modern society. | **K2** |
| **CO2** | Describe the evolution of cyber law in India and analyze the provisions of the Information Technology Act 2000 and related policies. | **K3** |
| **CO3** | Understand the role of Information Technology, its advantages and disadvantages, and evaluate its societal impacts including privacy and security issues. | **K4** |
| **CO4** | Analyze various cyber crimes and cyber frauds, identify preventive measures, and understand legal provisions related to cyber offences. | **K6** |
| **CO5** | Discuss investigation procedures, penalties under cyber laws, intellectual property rights, disaster recovery, and career opportunities in Information Technology. | **K6** |

**TEXT BOOKS:**

1. Cyber Law & Cyber Crimes (Information Technology Act, 2000) by Barkha Bhasin & Rama Mohan Rao.
2. Cyber Crimes & Digital Evidence – Indian Perspective by Asian School of Cyber Laws.
3. Commentary on Information Technology Act by Apar.

**REFERENCE LINK:**

- NPTEL Courses on Cyber Law and Cyber Security
- UGC e-Pathshala
- National Digital Library of India
- Introduction to Cyber Law – Legal Service India

| SECOND YEAR | INTERNSHIP | COURSE CODE: P25CSIS |
|---|---|---|
| SEMESTER- III | | CREDITS: 3 |

**COURSE OBJECTIVES:**

- Provide practical exposure to real-world working environments and industry practices.
- Apply theoretical knowledge to solve real-time problems in the chosen domain.
- Develop technical, analytical, and problem-solving skills through hands-on experience.
- Enhance professional communication, teamwork, and workplace ethics.
- Gain experience in project execution, documentation, and presentation of outcomes.

| S.NO | INTERNSHIP EVALUATION | MARKS |
|---|---|---|
| I | **Report of the Study**<br>• Introduction to organization and domain<br>• Objectives and scope of the internship<br>• Description of work carried out / project details<br>• Tools, technologies, and methodologies used<br>• Results, analysis, and outcomes<br>• Challenges faced and solutions implemented<br>• Learning outcomes and skills acquired<br>• Report presentation (format, clarity, documentation quality) | 80 |
| II | **Viva Voce**<br>• Understanding of the work/project done<br>• Conceptual clarity and technical knowledge<br>• Ability to explain tools and technologies used<br>• Problem-solving approach during internship<br>• Communication and presentation skills | 20 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand real-world organizational structure and professional work environment. | **K2** |
| **C02** | Apply theoretical knowledge to solve practical problems in the workplace. | **K3** |
| **C03** | Analyse tasks, challenges, and workflows in the chosen domain. | **K4** |
| **C04** | Develop and present a structured internship report with technical clarity. | **K6** |
| **C05** | Evaluate personal performance, skills gained, and professional growth during the internship. | **K5** |

# SEMESTER-IV

| SECOND YEAR | CORE COURSE–X | COURSE CODE: P25CSCC41 |
|---|---|---|
| SEMESTER- IV | BLOCKCHAIN TECHNOLOGY | CREDITS: 4 |

## COURSE OBJECTIVES:

- Understand the fundamentals of blockchain technology, decentralization, and distributed systems.
- Analyze cryptographic concepts, consensus algorithms, and smart contract mechanisms.
- Apply knowledge of Bitcoin architecture, transactions, mining, and blockchain networks.
- Evaluate Ethereum ecosystem, smart contracts, and decentralized application (DApp) development.
- Assess blockchain use cases, development tools, scalability, and security challenges.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **BLOCK CHAIN, DECENTRALIZATION** Block chain :The growth of block chain technology- Distributed systems - The history of block chain and Bit coin – Block chain - Consensus - CAP theorem and block chain. Decentralization: Decentralization using block chain - Methods of decentralization -Routes to decentralization – Block chain and full ecosystem decentralization - Pertinent terminology - Platforms for decentralization - Innovative trends. | 20 |
| II | **PUBLIC KEY CRYPTOGRAPHY, CONSENSUS ALGORITHMS AND SMART CONTRACTS:** Public Key Cryptography: Asymmetric cryptography - Cryptographic constructs and block chain technology. Consensus Algorithms: Introducing the consensus problem -Analysis and design - Classification - Algorithms - Choosing an algorithm. Smart Contracts: History - Definition - Ricardian contracts - Smart contract templates – Oracles - Deploying smart contracts - DAO | 20 |
| III | **BITCOIN** Bit coin: Bit coin—an overview- Cryptographic keys- Transactions- Block chain–Mining. Bit coin Network and Payments: The Bit coin network - Wallets – Bit coin payments -Innovation in Bit coin - Advanced protocols – Bit coin investment and buying and selling Bit coin. Bit coin Clients and APIs: Bit coin client installation - Experimenting further with bit coin-cli – Bit coin programming | 20 |
| IV | **ALTERNATIVECOINS** Alternative Coins: Theoretical foundations - Difficulty adjustment and retargeting algorithms – Bit coin limitations - Extended protocols on top of Bit coin -Development of alt coins. Ethereum: Ethereum – an overview - Ethereum network - Components of the Ethereum ecosystem – EthereumVirtual Machine (EVM) - Smart contracts. - Blocks and blockchain - Wallets and client - Nodes and miners - APIs, tools, and DApps - Supporting protocols - Programming languages. | 15 |
| V | **DEVELOPMENT TOOLSANDFRAMEWORKS, USECASES&SECURITY** Development Tools and Frameworks: Languages - Compilers - Tools and libraries - Frameworks - Contract development and deployment - Layout of a Solidity source code file - Solidity language. Use Cases: IoT – Government - Health -Finance – Media. Scalability and Other Challenges:Scalability-Privacy-Security-Otherchallenges | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand blockchain concepts, decentralization, and distributed ledger technologies. | **K2** |
| **C02** | Apply cryptographic techniques and consensus algorithms in blockchain systems. | **K3** |
| **C03** | Analyse Bitcoin architecture, transactions, and network operations. | **K4** |
| **C04** | Develop smart contracts and decentralized applications using Ethereum platform. | **K6** |
| **C05** | Evaluate blockchain applications, scalability, and security challenges in real-world scenarios. | **K5** |

**Text Books:**


1. Mastering Blockchain, Imran Bashir, Packt Publications, Third Edition, 2024.https://www.packtpub.com
2. Blockchain Basics A Non Technical Introduction in 25 Steps, Daniel Drescher, Apress Publications, Second Edition (Reprint), 2023.https://link.springer.com
3. Blockchain Technology Concepts and Applications, Arshdeep Bahga, Vijay Madisetti, Universities Press, Reprint Edition, 2024.https://www.universitiespress.com


**Referencebooks:**


1. Ethereum for Developers, Mayukh Mukhopadhyay, Apress Publications, Reprint Edition, 2022. https://link.springer.com
2. Hands On Blockchain with Hyperledger, Nitin Gaur, Packt Publications, Reprint Edition, 2021. https://www.packtpub.com
3. Introducing Ethereum and Solidity, Chris Dannen, Apress Publications, Reprint Edition, 2020. https://link.springer.com
4. Bitcoin and Cryptocurrency Technologies, Arvind Narayanan, Princeton University Press, Reprint Edition, 2021. https://press.princeton.edu
5. Blockchain Revolution, Don Tapscott, Alex Tapscott, Portfolio Publications, Reprint Edition, 2023. https://www.penguinrandomhouse.com

| SECOND YEAR | CORE COURSE–XI | COURSE CODE: P25CSCC42 |
|---|---|---|
| SEMESTER- IV | MACHINE LEARNING TECHNOLOGIES | CREDITS: 4 |

## COURSE OBJECTIVES:

- Understand the fundamentals of machine learning algorithms, data handling, and model selection techniques.
- Apply classification algorithms such as kNN, Naïve Bayes, and Decision Trees for predictive analysis.
- Analyse regression, neural networks, and support vector machines for forecasting and pattern recognition.
- Evaluate clustering techniques and model performance metrics for improving machine learning models.
- Develop deep learning models using perceptron's, CNNs, RNNs, and related architectures.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **Introducing Machine Learning:** The Origins of Machine Learning, Uses and Abuses of Machine Learning _ Basics of Machine Learning Algorithm Model Works - Steps to apply Machine Learning - Choosing a Machine Learning Algorithm - Using Machine Learning concepts. Managing and Understanding Data: Data Structures, Vectors And Factors: Lists, Data frames, Matrixes and arrays-Managing Data. | 20 |
| II | **Lazy Learning – Classification Using Nearest Neighbors:** The kNN Algorithm- Diagnosing Breast Cancer with the kNN Algorithm- Probabilistic Learning – Classification Using Naive Bayes: Basic concepts of Bayesian Methods- The Naïve Bayes Algorithm- Example – filtering Mobile Phone Spam with the Naive Bayes Algorithm. Divide and Conquer – Classification Using Decision Trees and Rules: Understanding Decision Trees- Example – Identifying Risky Bank Loans using C5.0 | 20 |
| III | **Forecasting Numeric Data – Regression Methods:** Understanding Regression- Example – Predicting Medical Expenses using Linear Regression- Understanding Regression Trees and Model Trees- Example – Estimating the Quality of Wines with Regression Trees and Model Trees. Black Box Methods Neural Networks and Support Vector Machines: Understanding Neural Networks, from Biological to Artificial Neurons, Activation Functions, Network Topology, Training Neural Networks with Back propagation | 20 |
| IV | **Finding Groups of Data – Clustering with K-Means:** Understanding Clustering- The k-means Algorithm for clustering- Finding teen market segments using k-means Clustering- Evaluating Model Performance: Measuring Performance for Classification- Beyond Accuracy – other Measures of Performance, Visualizing Performance Tradeoffs. Improving Model Performance: Tuning Stock Models for Better Performance-Using Caret for Automated Parameter Tuning- Creating a simple Tuned Model- Customizing the Tuning Process | 15 |
| V | **Introduction to Deep Learning:** Introduction to Deep Learning, Single Layer Perceptron Model (SLP), Multilayer Perceptron Model (MLP), Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Restricted Boltzmann Machines (RBMs). Convolution Neural Networks (CNNs): Structure and Properties of CNNs - Components of CNN Architectures- Convolution Layer, Pooling Layer, Rectified Linear Units (ReLU) | 15 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand machine learning concepts, data structures, and algorithm selection techniques. | **K2** |
| **C02** | Apply classification algorithms such as kNN, Naïve Bayes, and Decision Trees. | **K3** |
| **C03** | Analyse regression models, neural networks, and support vector machines for prediction tasks. | **K4** |
| **C04** | Evaluate clustering methods and optimize model performance using tuning techniques. | **K5** |
| **C05** | Develop deep learning architectures including CNNs, RNNs, and multilayer perceptron models. | **K6** |

**Text Books:**


1. Machine Learning with R, Brett Lantz, Packet Publications, Third Edition, 2023.https://www.packtpub.com
2. Deep Learning, Ian Good fellow, Yoshua Bagnio, Aaron Courville, MIT Press Publications, Reprint Edition, 2024.https://www.deeplearningbook.org
3. Hands On Machine Learning with Scikit Learn Keras and TensorFlow, Aurélien Géron, O'Reilly Publications, Third Edition, 2032.https://www.oreilly.com


**Referencebooks:**


1. Pattern Recognition and Machine Learning, Christopher M. Bishop, Springer Publications, Reprint Edition, 2021. https://link.springer.com
2. Machine Learning A Probabilistic Perspective, Kevin P. Murphy, MIT Press Publications, Reprint Edition, 2021. https://mitpress.mit.edu
3. Neural Networks and Deep Learning, Charu C. Aggarwal, Springer Publications, First Edition (Reprint), 2023. https://link.springer.com
4. Introduction to Machine Learning with Python, Andreas C. Müller, Sarah Guido, O'Reilly Publications, Reprint Edition, 2021. https://www.oreilly.com
5. Artificial Intelligence A Modern Approach, Stuart Russell, Peter Norvig, Pearson Publications, Fourth Edition, 2021.https://www.pearson.com

| SECOND YEAR | SOC-SKILL ORIENTED COURSE | COURSE CODE: P25CSSC1 |
|---|---|---|
| SEMESTER- IV | TECHNOLOGY INNOVATION AND SUSTAINABLEENTERPRISE | CREDITS: 4 |

## COURSE OBJECTIVES:

- Understand the concepts of innovation, knowledge management, and their role in organizational growth.
- Analyse technology transfer processes, joint ventures, and challenges in the Indian business environment.
- Apply web marketing principles, strategies, and legal considerations for online business development.
- Evaluate digital marketing techniques including online advertising, email marketing, and customer targeting.
- Assess enterprise resource planning systems and their applications in modern e-business environments.

| UNITS | CONTENTS | HOUR |
|---|---|---|
| I | **INNOVATION:** Need - Objectives of innovation - Technology innovation - its importance -Knowledge Management- need - Business strategies related to knowledgemanagement KnowledgeManagementApproaches – Transformationofanenterprise through Knowledge Management - Creating Knowledge Management System in Organization Establishing Knowledge Management through IT- Organizational culture for Knowledge Management - Future of Knowledge Management | 15 |
| II | **TECHNOLOGYTRANSFERANDJOINTVENTURES:** Policy, Procedure & Practices-India's Technology base and Capabilities-Preference of IndianTechnology majorConstraintsandproblems-Operationalconstraints-Problems in Indian Business Environment Problems in Finalization of Agreement- Major Problems in Technology transfer Collaboration Agreements. | 15 |
| III | **WEBMARKETING:** Meaning- Benefits of Web Marketing-Myths and Facts in Web Marketing Web Psychology: Understanding the Internet mind- The Internet and the Law: Copyright, Censorship, Privacy, Jurisdiction- Do's and Don'ts on Web | 10 |
| IV | **WEBMARKETINGSTRATEGIES:** Choosing the strategy- Online store fronts- Target Marketing Attracting Customers- Web Advertising - E-Mail Marketing-Instant market research - Securities Issues | 10 |
| V | **ENTERPRISE RESOURCE PLANNING:** The E- Business backbone -Meaning- ERP decision Enterprise Architecture Planning- ERP Implementation- The Future of ERP Applications- Procurement- Business Blueprint Planning. | 10 |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| **C01** | Understand innovation, knowledge management, and organizational transformation concepts. | **K2** |
| **C02** | Analyse technology transfer mechanisms and collaboration agreements in business environments. | **K4** |
| **C03** | Apply web marketing principles and strategies for online business promotion. | **K3** |
| **C04** | Evaluate digital marketing tools, customer engagement methods, and web security issues. | **K5** |
| **C05** | Assess ERP systems, implementation strategies, and enterprise architecture planning. | **K5** |

**Text Books:**

1. Knowledge Management in Theory and Practice, Kimiz Dalkir, MIT Press Publications, Fourth Edition, 2023. https://mitpress.mit.edu
2. Digital Marketing Strategy Implementation and Practice, Dave Chaffey, Fiona Ellis-Chadwick, Pearson Publications, Eighth Edition, 2022. https://www.pearson.com
3. Enterprise Resource Planning, Alexis Leon, McGraw Hill Publications, Fourth Edition (Reprint), 2021. https://www.mheducation.com

**Referencebooks:**

1. E Business and E Commerce Management, Dave Chaffey, Pearson Publications, Seventh Edition, 2022. https://www.pearson.com
2. ERP Demystified, Alexis Leon, McGraw Hill Publications, Reprint Edition, 2020. https://www.mheducation.com
3. Knowledge Management Systems and Processes, Irma Becerra-Fernandez, Rajiv Sabherwal, Routledge Publications, Second Edition, 2021.https://www.routledge.com
4. Principles of Marketing, Philip Kotler, Gary Armstrong, Pearson Publications, Eighteenth Edition, 2021. https://www.pearson.com
5. Technology Transfer and Innovation, Raghu Guard, Routledge Publications, Reprint Edition, 2020. https://www.routledge.com

| SECOND YEAR | DISSERTATION | COURSE CODE: P25CSPW |
|---|---|---|
| SEMESTER- IV | PROJECT | CREDITS: 5 |

**COURSE OBJECTIVES:**

- Apply theoretical and practical knowledge to solve real-world problems through project development.
- Develop research, analytical, and problem-solving skills in the chosen domain.
- Design and implement innovative solutions using appropriate tools and technologies.
- Enhance technical documentation, presentation, and project management skills.
- Strengthen professional ethics, teamwork, and independent learning abilities.

| S.NO | PROJECT EVALUATION |
|---|---|
| I | **Dissertation**<br>• Problem identification and objectives<br>• Literature survey and methodology<br>• System design and implementation<br>• Result analysis and validation<br>• Innovation and technical contribution<br>• Documentation quality and presentation |
| II | **Viva Voce**<br>• Understanding of project concepts<br>• Technical knowledge and implementation details<br>• Ability to explain methodology and outcomes<br>• Problem-solving and communication skills |

**COURSE OUTCOMES:**

Upon successful completion of this course the students would be able to:

| CO # | CO Statement | K-Level |
|---|---|---|
| CO1 | Understand and analyse real-world problems in the selected project domain. | K4 |
| CO2 | Apply appropriate tools, techniques, and technologies for project implementation. | K3 |
| CO3 | Design and develop innovative solutions to meet project objectives. | K6 |
| CO4 | Prepare and present a structured dissertation with technical and research clarity. | K6 |
| CO5 | Evaluate project outcomes, challenges, and future enhancements professionally. | K5 |

20/05/2026
R.M.LALLI, MCA.,M.Phil.,Ph.D.,
Associate Professor
Department of Computer Science,
Bharathidasan University,
Trichy - 620 023.

20/5/26
Dr. A. ADHISELVAM, Ph.D.
PROFESSOR AND HEAD
Department of Information Technology
Dr. N.G.P. Arts and Science College
Coimbatore - 641 048

T. MANIVANNAN,
Chairman - BOS,
E.G.S. Pillay Arts & Science College (A),
Nagapattinam - 611002.

20/5/26
Dr.P.PANIMALAR, M.C.A.,M.Phil.,S.E.T.,Ph.D.,
ASSOCIATE PROFESSOR OF COMPUTER SCIENCE
A.V.C. COLLEGE (AUTONOMOUS)
MANNAMPANDAL MAYILADUTHURAI-609 305.